

ATLIQ MEDIA · CODEBASICS RPC #21

Tamil Nadu Assembly Election 2026

AtliQ Media — Neutral Election Briefing

DOCUMENTATION BUNDLE

ECI public data only · 234 constituencies · 2021 vs 2026

Generated: May 28, 2026 · Descriptive analysis · No predictions, no causal claims

Documentation Bundle — Table of Contents

01	Project Overview	README
02	Hackathon Execution Plan	Execution Plan
03	Metric Definitions	Metric Definitions
04	Dashboard QA	Dashboard QA
05	Dashboard Reuse Guide	Reuse Guide
06	Chart Layout Audit	Chart Layout Audit
07	Video Narration Script	Narration Script
08	Visual & Reference Master	Visual Reference
09	Agent Instructions	Agent Instructions

This bundle compiles all design, methodology, QA, and reuse documentation for the Tamil Nadu Assembly 2026 election briefing submission.

1 · PROJECT OVERVIEW

Project Overview

Tamil Nadu Assembly 2026 — AtliQ Election Briefing

Interactive briefing dashboard for **AtliQ Media**: neutral analysis of Tamil Nadu Legislative Assembly results (2021 vs 2026), built from public Election Commission of India data only.

Codebasics Resume Project Challenge #21.

Disclaimer

Non-partisan data work. No political endorsement, predictions, or causal claims about why parties gained or lost seats.

Quick start

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
python scripts/build_processed_data.py

# API
uvicorn api.main:app --reload --port 8000

# Dashboard (separate terminal)
cd web && npm install && npm run dev
```

Open <http://localhost:5173>. The UI proxies `/api` to port 8000.

Production build (single port):

```
cd web && npm run build
uvicorn api.main:app --host 127.0.0.1 --port 8000
```

Share externally: `bash scripts/ngrok_share.sh`

What you see in the app

Section	Purpose
Editorial brief	Recommended 60-minute run-of-show and three core narrative threads
Margins	Fragmentation — margins, vote-share buckets, closest races
Seat flows	2021→2026 Sankey and incumbent retention
Statewide	KPIs, mosaic, seat tally, vote share
Explorer	Filters + sortable constituency table
Geography / Reserved / Ballots	Supporting views — regional vote share, reserved comparison, turnout
Deep insights	Electoral arithmetic — ENP, Pedersen volatility, Gallagher LSq, swing heatmap, representation gap, district churn
Sources	Definitions, data scope, caveats

Insight bullets on each page are computed from the same `data/processed/` tables as the charts.

Data & metrics

File	Role
<code>data/raw/tn_2021_results.csv</code>	2021 candidate-level results
<code>data/raw/tn_2026_results.csv</code>	2026 candidate-level results
<code>data/raw/constituency_master.csv</code>	234 ACs — district, macro-region, reserved
<code>data/processed/</code>	Joined AC tables, Sankey edges, regional aggregates

Join on `ac_number` (not constituency name). Definitions: `docs/METRIC_DEFINITIONS.md`.

Note: 2026 `turnout` is blank in the provided CSV. The pipeline reads `data/external/turnout_2026.csv` if present (template via `python scripts/fetch_eci_turnout_2026.py --template-only`); once filled from ECI Form-20, the `/depth` page automatically shows the Top-20 turnout-increase chart.

Headline numbers (this build)

- **108** seats — normalized winner TVK (2026)
- **163** normalized seat flips (2021→2026), ~70% of ACs
- Mean winning margin **11.8%** (2021) → **7.7%** (2026)

- **61** winners below **35%** of valid votes (2026)

Deep electoral arithmetic (Deep Insights page)

- **Effective Number of Parties** (Laakso–Taagepera): **3.74 → 4.29** statewide
- **Pedersen volatility**: **36.4** — academic literature flags 20+ as "very high"
- **Gallagher LSq disproportionality**: **15.2 → 8.7** — seats track votes more closely in 2026
- **132 of 234** ACs were multi-cornered races in 2026 (top-2 combined share < 70%)
- Definitions and constitutional context in [docs/METRIC_DEFINITIONS.md](#)

Deck, video, and exports

```
python scripts/build_all_deliverables.py      # processed data + deck + charts +  
walkthrough  
# Or step by step:  
python scripts/build_processed_data.py  
python scripts/build_presentation.py         # deck/TN_2026_AtliQ_Briefing.pptx (18 slides)  
python scripts/export_charts.py              # outputs/charts/  
python scripts/generate_walkthrough.py       # outputs/walkthrough/
```

Narration cues: [docs/VIDEO_NARRATION_SCRIPT.md](#). QA: [docs/DASHBOARD_QA.md](#). Reuse another election: [docs/DASHBOARD_REUSE.md](#).

Tests

```
cd web && npm run test:e2e  
python scripts/test_dashboard_api.py        # API on :8000
```

Repository map

Path	Purpose
<code>web/</code>	React + TypeScript + ECharts UI
<code>api/</code>	FastAPI over processed CSVs
<code>src/</code>	Metrics, normalization, charts
<code>scripts/</code>	Data build, exports, ngrok helper
<code>notebooks/</code>	Reproducible analysis
<code>config/election.json</code>	Jurisdiction metadata
<code>HACKATHON_EXECUTION_PLAN.md</code>	Internal execution checklist
<code>docs/DASHBOARD_QA.md</code>	Manual + E2E QA
<code>docs/DASHBOARD_REUSE.md</code>	Reuse pipeline for another election
<code>docs/VIDEO_NARRATION_SCRIPT.md</code>	Recording script aligned to routes
<code>deck/</code>	PowerPoint for video + RPC deck deliverable

Reproduce from a clean clone

```
rm -rf data/processed/*
python scripts/build_processed_data.py
uvicorn api.main:app --port 8000
```

2 · HACKATHON EXECUTION PLAN

Hackathon Execution Plan

TN Election RPC #21 — Master Execution Plan

Challenge: Codebasics Resume Project Challenge #21 — Decoding the 2026 Tamil Nadu Assembly Election

Role: Freelance data analyst for **AtliQ Media** (neutral, ECI-only TV storytelling)

Deadline: 28 May 2026, 11:59 PM IST

Prize: ₹10,000

Primary skill tested: Storytelling (70 pts) > analytical hygiene (30 pts)

This document is the **single source of truth** for execution. Keep it updated as decisions are made (headline, party mapping, turnout source).

0. Recommended winning strategy (data-informed)

0.1 What the starter data already shows

Quick profiling of the provided CSVs (join on `ac_number`):

Signal	2021	2026	Story use
Constituencies	234	234	Full coverage
ACs where winning party label changed	—	163 (~70%)	Q2 Flips — dominant narrative
TVK won (label TVK)	0	108 seats	Q3 Vote share / new entrant
Avg winning margin (% of valid votes)	11.8%	7.7%	Q6 — more fragmented races
Winner vote share >50%	(compute)	14	Q6 — few landslides
Winner vote share <35%	(compute)	61	Q6 — many pluralities
Turnout in CSV	234 ACs filled	0 (blank)	Q5 requires ECI scrape
Flips by region (2021 region tag)	—	South 38, Chennai 30, Kongu/Central/North ~26–27, Delta 15	Q1 Geographic

0.2 Recommended story package (best ROI for rubric)

Headline (example — refine after normalization):

"In 2026, Tamil Nadu's assembly became more fragmented: TVK took 108 seats, seven in ten constituencies changed hands, and average winning margins fell sharply versus 2021."

Pick these 3 research questions:

1. **Q6 — Margin of victory** (fragmentation, landslides vs pluralities)
2. **Q2 — Flip story** (163 seat changes; Sankey or alluvial)
3. **Q3 — Vote share / TVK** (where TVK won; vote share shifts by region)

Optional 4th thread for deck (not a 4th "question"): Q1 geographic — one regional heatmap/small multiples slide.

Do not lead with Q5 (turnout) unless you complete ECI turnout scrape early; it is extra work and not required for a winning story.

0.4 Deep electoral-science layer (extra credit, neutral)

In addition to the six research questions, we surface structural indices from comparative-politics literature on a dedicated **/deep** dashboard page and three deck slides (14–16):

- **Effective Number of Parties** (Laakso–Taagepera, 1979): vote-share weighted fragmentation

- **Pedersen volatility** (1979): net vote-share churn between elections
- **Gallagher LSq** (1991): seat-vote disproportionality under first-past-the-post
- **Representation gap** per party (seat share – vote share)
- **Anti-incumbency rate** by region / reserved category / party
- **Race competitiveness** buckets (multi-cornered / three-way / two-party)
- **District churn** with "full sweep" flag

All measures are descriptive, ECI-only, and add depth without changing the lead narrative. See [docs/METRIC_DEFINITIONS.md](#) for formulas and constitutional context.

0.3 Why this beats other combinations

- Hits **surprise + scale** reviewers want (108 TVK seats, 163 flips, margin collapse).
- Three charts that **look different** (distribution, flow, regional bars/maps).
- Fully **descriptive** — no causal claims about why DMK/AIADMK/TVK gained/lost.
- Avoids reserved-seat controversy (Q4) unless you have a crisp, neutral angle.

1. Non-negotiable rules (disqualification if broken)

- ☐ **Data:** ECI public data only (+ Codebasics starter CSVs). No exit polls, news, social media, opinion.
- ☐ **Tone:** No praise/criticism of party, leader, community, religion, region.
- ☐ **Claims:** No "because" explanations for outcomes; no predictions.
- ☐ **Visuals:** No leader photos; no party symbols/logos.
- ☐ **AI:** Allowed for scrape/clean/chart; you own story selection and neutral framing.

2. Deliverables checklist (strict priority)

Priority	Deliverable	Spec	Weight focus
1	Video	5–7 min, on camera, pitch to AtliQ content head; hook → 2–3 stories → editorial rec → limitations	20 pts storytelling + 20 video
2	Deck	8–10 slides: headline → stories → editorial recommendation → limitations	15 pts
3	GitHub	Public repo; Python/SQL notebook runs end-to-end; README with sources + reproduction	20 pts accuracy
4	Dashboard	Power BI / Tableau / Streamlit (optional)	10 pts (skip-scored if absent)
5	LinkedIn + RPC	Post with mandatory tags; submit post URL on RPC page	Required for evaluation

2.1 Editorial recommendation (deck + video)

Must answer for AtliQ: **What should the 60-minute show cover, in what order, and why?**

Example structure (neutral):

- Segment A (10 min): Statewide seat arithmetic — TVK entry, fragmentation metrics
- Segment B (15 min): Regional map — where flips clustered
- Segment C (10 min): Close races — margins under X%
- Segment D (5 min): Data caveats — live results vs Form-20, turnout sourcing

3. Repository structure (current)

```
Codebasics-Hackathon/
├── README.md
├── HACKATHON_EXECUTION_PLAN.md
├── config/election.json      # Jurisdiction metadata (+ /api/meta)
├── data/
│   ├── raw/                # tn_*_results.csv, constituency_master.csv
│   ├── external/           # Optional scraped 2026 turnout
│   └── processed/          # ac_comparison.csv, sankey_edges.csv, ...
├── notebooks/
│   └── 01_build_and_explore.ipynb
├── src/                    # metrics, advanced_metrics, party_normalize, charts,
presentation
├── api/                    # FastAPI (serves processed CSVs + /api/advanced/* +
optional web/dist)
├── web/                    # React + TypeScript + ECharts (primary dashboard, incl.
Deep insights page)
├── scripts/                # build_processed_data, build_presentation,
fetch_eci_turnout_2026, ngrok, ...
├── outputs/
│   ├── charts/             # Plotly HTML exports
│   ├── presentation_assets/ # PNGs embedded in deck
│   └── walkthrough/        # Video script, teleprompter, checklist
├── deck/                   # TN_2026_AtliQ_Briefing.pptx
├── dashboard/              # Legacy Streamlit (optional; not submission path)
└── docs/
    ├── METRIC_DEFINITIONS.md
    ├── DASHBOARD_QA.md
    ├── DASHBOARD_REUSE.md
    └── VIDEO_NARRATION_SCRIPT.md
```

4. Data dictionary & joins

4.1 Files

File	Rows	Key
<code>tn_2021_results.csv</code>	~4,232	<code>ac_number</code>
<code>tn_2026_results.csv</code>	~4,257	<code>ac_number</code>
<code>constituency_master.csv</code>	234	<code>ac_number</code>

4.2 Columns (results files)

- `constituency`, `ac_number`, `candidate`, `party`, `votes`, `turnout`, `reserved`, `region`
- Turnout:** populated in 2021; **empty in 2026** → scrape from `results.eci.gov.in/ResultAcGenMay2026` if needed

- **Name drift:** e.g. `Gummidipundi` vs `Gummidipoondi` — always join on `ac_number`, not name

4.3 Regions (editorial, not ECI)

Chennai Metro, North, Central, Kongu, Delta, South — use `constituency_master.region` as canonical.

4.4 Reserved seats

GEN 188 | SC 44 | ST 2 — same in 2021 and 2026.

5. Metric definitions (document in `docs/METRIC_DEFINITIONS.md`)

5.1 Valid votes per AC

```
valid_votes(ac) = SUM(votes) WHERE party ≠ 'NOTA'
```

(Confirm against ECI constituency PDF for 1–2 sample ACs; note definition in README.)

5.2 Winner

```
winner(ac) = candidate with MAX(votes) among non-NOTA candidates
```

Ties: flag manually (should be rare).

5.3 Vote share

```
vote_share(candidate) = votes / valid_votes(ac) * 100  
winner_share(ac) = votes(winner) / valid_votes(ac) * 100
```

5.4 Margin of victory

```
margin_votes = winner_votes - runner_up_votes  
margin_pct = margin_votes / valid_votes(ac) * 100
```

5.5 Flip

```
flip(ac) = winner_party_2026 ≠ winner_party_2021
```

After **party normalization** (Section 6). Report both raw-label flips (163) and normalized flips in limitations if they differ.

5.6 Seat counts by party

Count ACs where normalized winner party = X.

For “alliance” views, map parties to **formation** (DMK+, AIADMK+, etc.) only if you publish an explicit, source-backed mapping table.

6. Party normalization (critical for 2026)

2026 `party` field has long names and many `IND` rows. Build `party_normalize.csv`:

raw_party (examples)	normalized	notes
TVK	TVK	
DMK	DMK	
Anna Puratchi Thalaivar Amma Dravida Munnetra Kazhagam	AIADMK	verify spelling variants
All India Puratchi Thalaivar Makkal Munnettra Kazhagam	AIADMK	
INC	INC	
BJP	BJP	
NTK	NTK	
IND	IND	do not merge into majors without ECI “party” field proof

Process:

- `SELECT DISTINCT party, COUNT(*) FROM tn_2026 GROUP BY party ORDER BY count DESC`
- Manually map top 30 labels
- For winner stats, use **normalized** column only
- In README: “Independent candidates are labeled IND; vote-transfer claims are not made.”

7. Phase-by-phase execution (step-by-step)

PHASE 1 — Foundation (Days 1–2)

Step 1.1 — Fork starter data

- ☐ Copy `input_files_for_participants_rpc/data/*` → `data/raw/`
- ☐ Never edit raw files; all transforms write to `data/processed/`

Step 1.2 — Create GitHub repo

- ☐ Public repository
- ☐ Add `.gitignore` (exclude large caches, keep processed CSVs <10MB or use Git LFS if needed)
- ☐ Initial README: challenge name, disclaimer, folder map

Step 1.3 — Environment

- ☐ Python 3.11+ with `pandas`, `matplotlib` or `plotly`, `jupyter`
- ☐ Optional: `requests`, `beautifulsoup4` for turnout scrape
- ☐ `requirements.txt` pinned versions

Step 1.4 — Build AC-level tables

Notebook `01_build_ac_tables.ipynb`:

1. Load 2021 + 2026 + master
2. Per `ac_number`, compute: `valid_votes`, `winner`, `runner_up`, `winner_share`, `margin_pct`, `total_candidates`
3. Join master → `district`, `region`, `reserved`
4. Export `data/processed/ac_summary_2021.csv`, `ac_summary_2026.csv`
5. Export `data/processed/ac_comparison.csv` with columns:
`ac_number`, `region`, `reserved`, `winner_2021`, `winner_2026`, `flip`, `margin_2021`, `margin_2026`, `winner_share_2021`, `winner_share_2026`

Validation gates:

- ☐ 234 rows each year
- ☐ Sum of votes per AC matches manual check for AC 1 and AC 100
- ☐ No null `ac_number`

Step 1.5 — Party normalization

- ☐ Create `src/party_normalize.py` + `data/party_normalize.csv`
- ☐ Re-run AC tables with `winner_party_norm`

PHASE 2 — Story metrics (Days 3–4)

Notebook `02_story_metrics.ipynb`:

Q6 — Margins (do first)

- ☐ Statewide: mean/median margin 2021 vs 2026
- ☐ Count: `winner_share` > 50%, < 35% (both years)
- ☐ Chart: overlaid histogram or boxplot of `margin_pct`
- ☐ Table: top 10 closest races 2026 (neutral titles: "Smallest margin, AC name")

Q2 — Flips

- ☐ Count flips (normalized)
- ☐ Build transition matrix: `winner_2021 → winner_2026` (top 15 flows)
- ☐ Chart: Sankey (plotly) or alluvial — **no party colors that look like flags**; use neutral palette
- ☐ Optional: map flips per region (choropleth if you have GeoJSON; else bar chart by region)

Q3 — TVK / vote share

- ☐ TVK seats won by region
- ☐ Vote share by party statewide (2021 vs 2026) — bar chart
- ☐ Regional small multiples: DMK, AIADMK, TVK share by region
- ☐ **Do not claim** “TVK took votes from X” — say “TVK’s vote share was Y% in region Z; DMK share changed from A to B” (descriptive only)

Q1 — Geographic (one slide)

- ☐ Seats won by major normalized parties per region, 2021 vs 2026 side-by-side stacked bar

Q5 — Turnout (only if time)

- ☐ Scrape 2026 turnout per AC from ECI portal
- ☐ Save `data/external/turnout_2026.csv` with source URL + scrape date
- ☐ Merge; compute delta vs 2021; top 20 increases
- ☐ If skipped: one limitations bullet in deck

PHASE 3 — Visual design system (Day 5)

- ☐ **Palette:** Colorblind-safe (e.g. Okabe-Ito); avoid saffron/green combos that imply parties
- ☐ **Fonts:** Single sans-serif; large axis labels for TV
- ☐ **Titles:** Fact-only — “Average winning margin fell from 11.8% to 7.7%” not “DMK crushed”
- ☐ Export charts at 1920×1080 min → `outputs/charts/`
- ☐ Chart inventory table:

Chart ID	Question	File	Deck slide
C1	Q6	margin_hist.png	3
C2	Q2	sankey_flips.png	4
C3	Q3	tvk_by_region.png	5
C4	Q1	seats_by_region.png	6

PHASE 4 — Deck (Days 6–7)

18 slides (current build) — generated by `python scripts/build_presentation.py` :

1. Title — Headline finding
 2. The challenge — RPC #21 problem statement
 3. Our solution — Pipeline → three narratives → dashboard
 4. Technology stack
 5. Data & methodology
 6. Research questions — coverage map (Q1–Q6) + electoral arithmetic note
 7. **Q6 LEAD** — Margin of victory
 8. **Q2 LEAD** — Seat churn at scale
 9. **Q3 LEAD** — Vote share statewide
 0. Q3 — Vote share by region
 1. Q1 — Geographic pattern
 2. Q4 — Reserved seats (GEN / SC / ST)
 3. Q5 — Turnout
 4. **Deep insight** — Electoral arithmetic (ENP / Pedersen / Gallagher)
 5. **Deep insight** — Regional swing heatmap
 6. **Deep insight** — Representation gap (vote vs seat share)
 7. Editorial recommendation — 60-minute show
 8. Limitations & reproducibility
- ☐ Export PDF
 - ☐ Place in `deck/` and link from README
-

PHASE 5 — Video (Days 8–9)

Step 5.1 — Script (600–900 words for 5–7 min)

Structure:

1. **Hook (30s):** Headline + one surprising number
 2. **Context (45s):** AtliQ brief — neutral facts only
 3. **Story 1 (90s):** Margins — show C1
 4. **Story 2 (90s):** Flips — show C2
 5. **Story 3 (90s):** TVK / regions — show C3–C4
 6. **Recommendation (60s):** Show segment plan
 7. **Limitations (30s):** Honest caveats
 8. **Close (15s):** Thank you
- ☐ Bullet script, not word-for-word essay — avoids “reading script” penalty

- ☐ Rehearse 3 times; record face + screen share

Step 5.2 — Production

- ☐ External mic; quiet room
- ☐ 1080p screen recording; charts legible
- ☐ Upload YouTube **unlisted** or Google Drive public link

PHASE 6 — Dashboard (Days 9–10, optional)

If built — **Streamlit recommended** for speed:

- ☐ Page 1: Filters (region, reserved, flip yes/no)
- ☐ KPI cards: total flips, TVK seats, avg margin 2026
- ☐ Table: AC list sortable by margin
- ☐ Do not duplicate every deck chart — exploration only

PHASE 7 — QA & submission (Days 10–12)

Step 7.1 — Reproducibility

- ☐ Fresh clone → `pip install -r requirements.txt` → run notebooks top-to-bottom
- ☐ All output numbers match deck/video (spreadsheet cross-check)

Step 7.2 — Tone audit

For every slide title and chart title, ask:

"Would a supporter of any party find this offensive or partisan?"

- ☐ Replace loaded words: "wave", "sweep", "rebel", "vs battleground"

Step 7.3 — Reviewer simulation

- ☐ Watch video once without deck — write one-sentence headline
- ☐ If unclear, re-cut opening 60 seconds

Step 7.4 — LinkedIn post

Include:

- ☐ Link to GitHub
- ☐ Link to video
- ☐ Tags: `@codebasics`, Hemanand Vadivel, Dhaval Patel, Government of Tamil Nadu
- ☐ Hashtags: `#ResumeProjectChallenge #Codebasics #DataAnalytics #TamilNaduElection2026`

Step 7.5 — RPC portal

- ☐ Submit LinkedIn post URL on Resume Project Challenge page before deadline

8. Evaluation rubric map (100 points)

Criterion	Pts	How this plan addresses it
Story selection & framing	20	Single headline; 3 linked questions; regional slide
Video performance	20	Script structure; rehearsal; hook + recommendation
Deck narrative arc	15	8-slide arc with editorial rec
Data accuracy & reproducibility	20	Metric doc, validation gates, public GitHub
Dashboard	10	Optional Streamlit
Non-partisanship	10	Tone audit checklist
Limitations honesty	5	Dedicated slide + video beat

9. Timeline (working backward from 28 May 2026)

Dates	Milestone
Day 1–2	Repo + AC tables + party normalization
Day 3–4	All core metrics + charts C1–C4
Day 5	Visual polish + chart exports
Day 6–7	Deck v1 → final PDF
Day 8–9	Video script, record, upload
Day 10	Dashboard (optional) + QA
Day 11–12	Tone audit, LinkedIn, RPC submit
28 May	Buffer / only fixes

10. Risk register

Risk	Mitigation
Party label chaos in 2026	Normalization table + document uncertainty
163 raw flips include alliance artifacts	Report normalized flips; footnote methodology
Turnout blank in 2026 CSV	Skip Q5 or scrape early with dated external file
Constituency name mismatch	Join only on <code>ac_number</code>
Over-claiming TVK "vote source"	Descriptive share changes only
Disqualification on tone	Tone audit + no causal language

11. Quick reference — file paths in workspace

```
input_files_for_participants_rpc/  
├─ Readme-first-brief.docx  
├─ research_questions_and_recommendations.docx  
├─ How_to_Submit.docx  
├─ metadata.txt  
└─ data/  
    ├─ tn_2021_results.csv  
    ├─ tn_2026_results.csv  
    └─ constituency_master.csv
```

12. Decision log (update as you go)

Date	Decision	Rationale
	Headline text	
	3 questions chosen	Q6, Q2, Q3 recommended
	Turnout: include Y/N	
	Dashboard tool	

Last updated: 28 May 2026 — added deep electoral-arithmetic layer (ENP, Pedersen, Gallagher), new `/deep` page, 18-slide deck.

3 · METRIC DEFINITIONS

Metric Definitions

Metric definitions — TN Assembly 2021 vs 2026

Use these consistently across **notebook**, **dashboard**, **deck**, and **video**.

Valid votes (per AC, per year)

```
valid_votes = SUM(votes) WHERE UPPER(party) ≠ 'NOTA'
```

Winner

Candidate with maximum `votes` among non-NOTA rows for that `ac_number`.

Vote share (%)

```
vote_share = 100 * candidate_votes / valid_votes  
winner_share = vote share of winner
```

Margin of victory (%)

```
margin_pct = 100 * (winner_votes - runner_up_votes) / valid_votes
```

Flip

```
flip = (winner_party_norm_2026 ≠ winner_party_norm_2021)
```

Requires `party_normalize.csv`. Report raw-label flip count separately if it differs.

Seat count

Count of ACs where `winner_party_norm` equals party X. Alliances only if explicit mapping table is published in README.

Join key

Always `ac_number` (1-234). Do not join on `constituency` name alone.

Turnout

- 2021: from starter CSV `turnout` column (constituency-level, repeated per row).
- 2026: **not in starter pack** — source from ECI portal; record scrape date in `data/external/`.

Advanced electoral indices (Deep Insights page)

All indices are descriptive measures from comparative-politics literature.

Effective Number of Parties (ENP) — Laakso & Taagepera (1979)

$$\text{ENP} = 1 / \sum p_i^2$$

where `pi` is each party's share of valid votes (0–1). ENP = 2 in a pure two-party contest; values of 3–4 indicate moderate fragmentation; > 4 is high.

Pedersen index of volatility — Pedersen (1979)

$$V = \frac{1}{2} \sum |p_i(t) - p_i(t-1)|$$

where shares are in percentage points. The index measures net vote-share churn between two elections. Values above ~20 are considered very high in stable democracies; 36 (TN 2021→2026) indicates structural realignment.

Gallagher Least-Squares disproportionality (LSq) — Gallagher (1991)

$$\text{LSq} = \sqrt{\frac{1}{2} \sum (s_i - v_i)^2}$$

where `si` is a party's seat share and `vi` is its vote share. Lower values mean seats match votes more closely. Compared with the older Loosemore–Hanby index, LSq weights large discrepancies more heavily — closer to international standard.

Representation gap (per party)

$$\text{gap_pp} = \text{seat_share_pct} - \text{vote_share_pct}$$

Positive values indicate first-past-the-post amplification of a party's vote into seats; negative values indicate under-representation.

Anti-incumbency rate

$$\text{anti_incumbency_pct} = 100 \times \sum \text{flip} / \text{total_seats}$$

within a slice (region, reserved category, or party 2021). Descriptive only — counts seats where the 2021 winning party did not retain the seat in 2026. No claim about voter motivation.

Race competitiveness

For each AC: combined vote share of top two candidates.

```
top2 = (votes_winner + votes_runner_up) / valid_votes * 100
```

Buckets used:

- **Multi-cornered** : $\text{top2} < 70\%$
- **Three-way** : $70\% \leq \text{top2} < 85\%$
- **Two-party** : $\text{top2} \geq 85\%$

District "full sweep"

A district where every assembly constituency it contains changed normalised winner between 2021 and 2026.

Constitutional context (descriptive)

- **Total seats:** 234 (TN Legislative Assembly, fixed under the Constitution and the Delimitation Act).
- **Reserved seats:** 44 SC and 2 ST in the current delimitation (Articles 332/334 of the Constitution; Delimitation Order 2008). The reserved set is identical across 2021 and 2026 in the starter file.
- **Voting system:** Simple-majority first-past-the-post. Winners can be elected on a plurality (< 50%) of valid votes — this is by design and not a defect.
- **NOTA:** A constitutional option since SC ruling 2013 (PUCL vs Union of India). Excluded from valid votes in margin and share computations to match ECI's "valid vote" reporting.

Sources

- Codebasics: `tn_2021_results.csv`, `tn_2026_results.csv`, `constituency_master.csv`
- ECI: <https://results.eci.gov.in/>
- 2021 lineage: Trivedi Centre / ECI (per `metadata.txt`)
- Indices: Laakso & Taagepera (1979), Pedersen (1979), Gallagher (1991)

4 · DASHBOARD QA

Dashboard QA

Dashboard QA checklist

Stack: **React** (`web/`) + **FastAPI** (`api/`). Legacy Streamlit under `dashboard/` is not used for submission.

Playwright E2E (recommended)

```
cd web
npm install
npm run test:e2e          # uses system Google Chrome (channel: chrome)
```

Expect **56 tests** (navigation, charts, filters, viewport, API contract). Re-run after UI changes and update this doc with the latest pass count.

Report: `cd web && npm run test:e2e:report`

Automated API smoke test

```
source .venv/bin/activate
uvicorn api.main:app --port 8000 &
python scripts/test_dashboard_api.py
```

Manual UI checklist (each nav page)

Route	Nav label	Fit / scroll	Interactivity	Notes
/	Editorial brief	Scroll	Links to story pages	60-min run-of-show, three core threads, limitations
/margins	Margins	Viewport charts	dataZoom on scatter	Lead story — fragmentation
/flows	Seat flows	Viewport charts	"All flows" toggle	Lead story — 2021→2026 transitions
/overview	Statewide	Viewport charts	Year 2021/2026; region chips; tile → detail	Lead story — KPIs, mosaic, vote share
/explorer	Explorer	Scroll (table)	Filters; scatter click	Producer drill-down
/geography	Geography	Viewport charts	Map mode; year on stacks	Supporting — regional pattern
/reserved	Reserved seats	Viewport charts	—	Supporting — GEN/SC/ST counts
/depth	Ballots & turnout	Viewport charts	—	2021 turnout only; limitation banner
/deep	Deep insights	Scroll (5 panels)	—	Electoral arithmetic — ENP, Pedersen, Gallagher, swing heatmap, representation gap, district churn
/methods	Sources	Scroll	—	Definitions, data scope, caveats

Insight panels show **page-specific bullets** from `/api/insights?year=` (no Q1–Q6 labels in the UI).

Year filter regression

Playwright: `year toggle updates KPIs, mosaic, and bar charts` on `/overview` — 2021 shows DMK 133 seats, zero TVK tiles; 2026 shows TVK tiles.

Manual: toggle 2021 on Statewide → mosaic mostly DMK/AIADMK, KPIs show DMK 133.

Reuse for another election

See `docs/DASHBOARD_REUSE.md` and `config/election.json`. No in-app CSV upload — swap `data/raw/` files and rebuild processed CSVs.

Viz best practices applied

- **No 3D pie** — vote share uses horizontal bars.
- **Equal-weight tiles** — NPR-style mosaic (234 = 18×13).
- **Year toggle** — 2021 | 2026 on Statewide + Geography; KPIs/mosaic/bars follow year.
- **Sankey** for seat flows (AP VoteCast pattern).
- **Descriptive copy** — no causal or predictive language.
- **Charts resize** — `EChart` uses `ResizeObserver` when `height="fill"`.
- **Party colors** documented in `web/src/lib/colors.ts`.

Known limits

- Explorer, Sources, and Deep insights scroll by design (long tables / multi-panel layout).
- Sankey with “all flows” can be dense on small laptops — use default top flows for recording.
- 2026 turnout not in starter CSV — optional ECI scrape to `data/external/`.
- District map is **2026-only**; flip mode is 2021→2026.
- Mobile: hamburger nav; pages scroll vertically on small screens.
- Swing heatmap colors are diverging (red = loss, green = gain) — labels include sign for accessibility.

Local run modes

Mode	Command	URL
Dev (hot reload)	<code>uvicorn api.main:app --port 8000</code> + <code>cd web && npm run dev</code>	http://127.0.0.1:5173
Single port	<code>cd web && npm run build</code> then <code>uvicorn api.main:app --port 8000</code>	http://127.0.0.1:8000
Share	<code>bash scripts/ngrok_share.sh</code>	ngrok URL → :8000

Dashboard Reuse Guide

Reusing the dashboard for another election

What is reusable today

Layer	Reusable?	How
Pipeline	Yes	<code>scripts/build_processed_data.py</code> + <code>src/metrics.py</code> — swap CSVs in <code>data/raw/</code> , keep <code>ac_number</code> join
Advanced metrics	Yes	<code>src/advanced_metrics.py</code> — ENP, Pedersen, Gallagher LSq, swing, anti-incumbency, race competitiveness all compute generically from any two-election dataset with normalized parties
API	Yes	FastAPI reads <code>data/processed/*.csv</code> — logic is not TN-specific; new <code>/api/advanced/*</code> endpoints follow the same pattern
React app	Mostly	Party colors in <code>web/src/lib/colors.ts</code> ; copy in page components; metadata in <code>config/election.json</code>
Deep insights page	Yes	<code>web/src/pages/DeepInsights.tsx</code> is data-driven — only the glossary block carries jurisdiction-neutral copy
Maps	Partial	TN district GeoJSON + <code>district_to_geo.json</code> — replace for other states
Insights text	Generated	<code>api/services.py</code> <code>get_insights()</code> and <code>_deep_bullets()</code> — recomputed from processed tables

What is NOT in the starter pack (cannot add without new data)

- Polling **booth**-level results
- Voter **age** / demographic tables
- **2026 turnout** (blank in provided CSV; 2021 turnout used on Ballots & turnout page)

Steps for another state or India-wide (next cycle)

1. **Prepare ECI CSVs** with the same logical columns: `ac_number`, `candidate`, `party`, `votes`, `constituency`, optional `turnout`, `region`, `reserved`.
2. Copy files into `data/raw/` (or update paths in `config/election.json`).

3. Build `data/processed/party_normalize.csv` for new party label variants.

4. Run:

```
source .venv/bin/activate
python scripts/build_processed_data.py
python scripts/build_district_map_geo.py # after adding GeoJSON + district_to_geo.json
```

5. Update `web/src/lib/colors.ts` for any new major parties.

6. Replace `web/public/geo/tn_districts_map.geojson` (or add a state-specific map component).

7. Start API + web:

```
uvicorn api.main:app --reload --port 8000
cd web && npm run dev
```

Config-driven UI

Feature	Status
Election metadata (<code>config/election.json</code>)	Done — jurisdiction, years, regions, file paths
<code>GET /api/meta</code>	Done — returns <code>election.json</code>
Dynamic titles from API in React	Partial — sidebar copy is still TN-specific
Upload CSV in browser	Not built — post-hackathon
Multi-state GeoJSON picker	Not built — one map per deployment

Research threads → dashboard pages (internal mapping)

Thread	Dashboard pages
Geographic patterns	Geography, Statewide mosaic
Seat flips	Seat flows, Statewide KPIs
Vote share / new entrant	Statewide, Explorer
Reserved seats	Reserved
Turnout	Ballots & turnout (2021 only in starter data)
Margins / fragmentation	Margins, Statewide KPIs
Electoral arithmetic (extra)	Deep insights — ENP, Pedersen, Gallagher, swing heatmap

Insights panels use `/api/insights?year=` — bullets only, keyed by page (`overview`, `flows`, `margins`, `deep`, etc.).

Advanced metric endpoints (regenerated automatically)

All `/api/advanced/*` endpoints read from CSVs written by `src/advanced_metrics.build_advanced_tables`:

- `/api/advanced/summary` — ENP / Pedersen / Gallagher / incumbent-loss headline indices
- `/api/advanced/enp-by-region`
- `/api/advanced/pedersen-by-region`
- `/api/advanced/swing-by-region` (party × region matrix)
- `/api/advanced/anti-incumbency`
- `/api/advanced/representation-gap` (vote share vs seat share)
- `/api/advanced/district-flips`
- `/api/advanced/bellwethers`
- `/api/advanced/race-types`

All require: (a) normalized party labels, (b) `ac_comparison.csv` joined across two elections, (c) `region` and `reserved` columns from the master.

Chart Layout Audit

Chart layout audit — all pages

Checklist for **overlap**, **right-edge trim**, and **bottom trim**.

Implementation: `web/src/lib/panelLayout.ts`, `web/src/components/ChartPanel.tsx`, `web/src/charts/chartLayout.ts`.

Layout primitive	Use when
<code>panelHeightMd</code> / <code>Lg</code> / <code>XL</code>	Single chart or map panel (fixed px height)
<code>chartStack</code>	2 charts in right column (Statewide, Geography, Margins)
<code>chartPageStack</code>	Vertical stack (Seat flows)
<code>pageChartGrid2</code>	2 columns of equal panels (Ballots & turnout, Reserved, Explorer)
<code>pageChartGridSplit</code>	Map/mosaic left + <code>chartStack</code> right
<code>ChartPanel</code>	Standard chart wrapper (<code>height</code> : <code>md</code> <code>lg</code> <code>xl</code> <code>fill</code>)
<code>chartAreaCompact</code>	ECharts slot — no extra min-height

0. Editorial brief (/)

No charts — text and run-of-show links only.

1. Statewide (/overview)

Chart	Layout	Status
KPI strip	Grid	✓
Mosaic	<code>panelHeightLg</code> + <code>chartAreaMosaic</code>	✓
Seat tally	<code>chartStack</code> → <code>ChartPanel</code> fill	✓
Vote share	<code>chartStack</code> → <code>ChartPanel</code> fill	✓

2. Seat flows (/flows)

Chart	Layout	Status
Sankey	<code>chartPageStack</code> → <code>ChartPanel</code> xl	✓
Retention	<code>chartPageStack</code> → <code>ChartPanel</code> md	✓

3. Geography (/geography)

Chart	Layout	Status
District map	<code>panelHeightLg</code>	✓
Regional stack	<code>chartStack</code> → <code>ChartPanel</code> fill	✓
Flip rate	<code>chartStack</code> → <code>ChartPanel</code> fill	✓

4. Margins (/margins)

Chart	Layout	Status
Margin scatter	<code>panelHeightLg</code> + <code>chartArea</code>	✓
Vote share buckets	<code>chartStack</code> → <code>ChartPanel</code> fill	✓
Closest races table	<code>chartStack</code> → <code>tableScroll</code>	✓

5. Reserved (/reserved)

Chart	Layout	Status
Flip by category	<code>ChartPanel</code> md	✓
Party × reserved table	<code>panelHeightMd</code> + <code>tableScroll</code>	✓

6. Ballots & turnout (/depth)

Chart	Layout	Status
Turnout by region	<code>pageChartGrid2</code> → <code>ChartPanel</code> md	✓
Turnout vs margin	<code>ChartPanel</code> md	✓
Candidate buckets	<code>ChartPanel</code> md	✓
NOTA top 12	<code>ChartPanel</code> md	✓

7. Explorer (/explorer)

Chart	Layout	Status
Filtered seat tally	<code>pageChartGrid2</code> → <code>ChartPanel</code> md	✓
Margin scatter	<code>ChartPanel</code> md	✓
AC table	<code>panelHeightMd</code> + <code>tableScroll</code>	✓

8. Deep insights (/deep)

Chart	Layout	Status
KPI strip (5 indices)	Grid	✓
ENP by region	<code>ChartPanel</code> lg	✓
Pedersen by region	<code>ChartPanel</code> lg	✓
Swing heatmap (party × region)	<code>ChartPanel</code> xl	✓
Anti-incumbency by region	<code>ChartPanel</code> lg	✓
Vote share vs seat share	<code>ChartPanel</code> lg	✓
Race competitiveness buckets	<code>ChartPanel</code> md	✓
District churn table	<code>Panel</code> + <code>tableScroll</code>	✓
Glossary	<code>panel</code> text block	✓

Page scrolls intentionally (8+ panels). Heatmap label color flips at $v_{max} \times 0.4$ for legibility.

9. Sources (`/methods`)

No charts — text only.

QA (all pages, ~3 min)

1. Hard-refresh (`Cmd+Shift+R`).
 2. Open each nav item; confirm **no chart drawn on top of another**.
 3. Check **right edge** of every bar/Sankey label.
 4. `cd web && npm run test:e2e`
-

Related

- `docs/DASHBOARD_QA.md`
- `docs/VIDEO_NARRATION_SCRIPT.md`

Video Narration Script

Video narration script — TN 2026 dashboard

Use this document as a **teleprompter-style guide** while recording your 5–7 minute AtliQ pitch. Numbers below come from the processed ECI starter files (`data/processed/`, normalized party labels). Re-run `python scripts/build_processed_data.py` if your pipeline changes.

Tone: Neutral, descriptive, ECI-only. Say what the data **shows** — not why voters behaved, not who should win.

Avoid: "because," "due to," "wave," "rejected," "mandate for/against," leader names, party symbols, predictions.

App routes: Editorial brief `/` · Statewide `/overview` · Margins `/margins` · Seat flows `/flows` · Geography `/geography` · Reserved `/reserved` · Ballots & turnout `/depth` · Deep insights `/deep` · Explorer `/explorer` · Sources `/methods`

Run locally: `uvicorn api.main:app --port 8000` after `cd web && npm run build`, or dev mode on <http://127.0.0.1:5173> with API on :8000.

Suggested video arc (≈7 minutes)

Segment	Time	Dashboard	Purpose
Hook + headline	0:00–0:45	Statewide /overview (2026)	Statewide story in one sentence
Seat flows	0:45–2:00	/flows	Where seats moved
Margins	2:00–3:00	/margins	Fragmentation
Vote share / TVK	3:00–3:45	/overview (bars + mosaic)	New entrant scale
Geography	3:45–4:30	/geography	Regional pattern
Deep insights	4:30–5:15	/deep	Electoral arithmetic — ENP, Pedersen, representation gap
Limitations	5:15–5:45	/methods (+ /depth if needed)	Data scope, no booth/age, 2026 turnout
Editorial recommendation	5:45–6:45	/ or deck slide 17	What AtliQ should air, in what order

Opening hook (on Statewide or on camera)

*“For a neutral sixty-minute briefing, the 2026 Tamil Nadu Assembly results in the ECI files show a more fragmented assembly than 2021: **TVK won 108 of 234 seats, 163 constituencies—about seven in ten—changed normalized winner, and the average winning margin fell from 11.8% to 7.7%.** I’ll walk through seat flows, margins, and geography using only public ECI data.”*

Page 0 — Editorial brief ([/](#))

Page purpose: Recommended 60-minute run-of-show for producers — segment order, minutes, and links to supporting views.

Script:

"Before the charts: here's how I'd structure the hour — open with statewide arithmetic, then regional patterns, then churn and close races, and end with sources and caveats. The three lead threads are margins, seat flows, and where TVK won. Everything below is descriptive ECI data only."

Page 1 — Statewide ([/overview](#))

Page purpose: One-screen statewide summary — seats, vote share, and equal-weight map of all 234 ACs.

Controls to demo: Year toggle **2021** | **2026**; region chips (optional filter).

Key insights panel

What it is: Auto-generated bullets from [/api/insights?year=](#) for this page only.

Script:

"The insights panel restates the same numbers as the charts—so the briefing stays consistent. Toggle the year to compare 2021 and 2026 without changing the story's neutrality."

KPI strip (4 cards)

What it shows: Headline counts for the **selected year**.

Year	Cards	Numbers to cite
2026	TVK seats, Flips, Avg margin, Under 35% share	108 / 234 · 163 (69.7%) · 7.7% (vs 11.8% in 2021) · 61 plurality wins
2021	Leading party seats, Runner-up, Avg margin, Under 35% share	DMK 133 · AIADMK 66 · 11.8% · 2 under 35%

Script (2026):

*"In 2026, TVK is the largest seat block in this file with **108 seats**. **163 constituencies flipped** on normalized party labels—that's **69.7%** of the assembly. The average winning margin is **7.7%**, down from **11.8%** in 2021. **61 winners** took under **35%** of valid votes—a plurality, not a majority of ballots."*

Script (2021 — after toggling year):

*"In 2021, **DMK led with 133 seats**, AIADMK had **66**, and the average margin was **11.8%**. Only **two** winners were under 35% vote share—so 2026 is visibly more fragmented on this metric."*

Hackathon link: Q2 (flips), Q3 (TVK seats), Q6 (margins, under-35%).

Chart: Mosaic — 234 constituencies

What it shows: One tile per assembly constituency (18×13 grid). Color = **normalized winner for selected year**. Equal area—no geographic distortion.

How to read: Dense yellow in 2026 = TVK breadth; red/blue dominant in 2021 = DMK/AIADMK era.

Script (2026):

"This mosaic treats every constituency equally—like an NPR-style tile map. Yellow is TVK, red DMK, blue AIADMK. You can see how widespread TVK's 108 wins are without exaggerating Chennai or Kongu on a geographic map."

Script (2021):

"Switching to 2021, the mosaic is mostly red and blue—DMK and AIADMK—with no TVK tiles, since TVK did not win seats on this label in 2021."

Interaction: Click a tile → AC detail card (name, district, 2021 vs 2026 winner, margins).

Hackathon link: Q1 (pattern at a glance), Q3 (TVK geography).

Chart: Seat tally (horizontal bars)

What it shows: Number of assembly seats won per normalized party for the **selected year**.

Script (2026):

"The seat tally confirms the mosaic: TVK has the most seats in 2026, followed by DMK and AIADMK on normalized labels. Bar length is seat count—not vote share."

Script (2021):

*"In 2021, DMK's bar is longest at **133 seats**, then AIADMK at **66**."*

Hackathon link: Q2, Q3.

Chart: Vote share (horizontal bars)

What it shows: Each party's share of **statewide valid votes** (NOTA excluded) for the selected year.

Script (2026):

*"Vote share is a different lens from seats. TVK has about **35%** of valid votes statewide in this file—while seat counts depend on where that vote was distributed across 234 constituencies."*

Script (2021):

*"In 2021, DMK's statewide vote share was about **38%**—similar magnitude, but the seat outcome was DMK-heavy because of how votes clustered."*

Hackathon link: Q3.

Page 2 — Seat flows (/flows)

Page purpose: How **2021 winning parties** map to **2026 winners**—seat counts only, not vote transfer.

Control: "All flows" checkbox (default: top paths).

Key insights panel

Script:

*"Seat flows answer hackathon question two: where did seats go between elections? The largest single band in the top flows is **2021 DMK to 2026 TVK—65 constituencies** in this file."*

Chart: Seat flow Sankey (top)

What it shows: Bands between nodes labeled **Party (2021)** and **Party (2026)**. **Band width = number of constituencies** that took that path.

Top flows to cite:

From (2021)	To (2026)	Seats
DMK	TVK	65
DMK	DMK	40
AIADMK	TVK	26
AIADMK	AIADMK	22
DMK	AIADMK	22

Script:

*“Read the Sankey left to right: 2021 winner on the left, 2026 winner on the right. The thickest band here is **DMK 2021 to TVK 2026—65 seats**. That’s descriptive flow—**not** voters ‘switching party’ as a one-to-one transfer. Smaller bands show seats DMK or AIADMK held, or moved to each other.”*

Hackathon link: Q2.

Chart: Incumbent retention (bottom)

What it shows: For each party that **won seats in 2021**, what **percentage of those same constituencies** it **still won in 2026**. Label format: retained / held (e.g. **40/133** for DMK).

How to read: Low retention = most 2021 wins for that party flipped to another 2026 winner. High retention = defended turf.

Numbers to cite:

Party	Retained / held 2021	Retention %
DMK	40 / 133	30.1%
AIADMK	22 / 66	33.3%
INC	4 / 18	22.2%
CPI	2 / 2	100%

Script:

*"Retention answers a simpler question than the Sankey: of the seats a party won in 2021, how many did it still win in 2026? **DMK kept 40 of 133**—about **30%** retention. **AIADMK kept 22 of 66**. TVK doesn't appear here as a 2021 incumbent because it had no 2021 wins on this label. This chart does **not** show who took lost seats—that's the Sankey above."*

Hackathon link: Q2.

Page 3 — Geography (/geography)

Page purpose: Regional and district views of **where** outcomes and flips cluster.

Controls: Year toggle (regional chart only); map mode **2026 winner** vs **flip rate**.

Note when year = 2021: District map still shows **2026**; banner on screen explains this.

Key insights panel

Script (2026):

*"Geography addresses question one. **Chennai Metro** has the highest flip rate in this file—**93.8%** of its 32 ACs changed normalized winner. **Delta** is lowest at **45.5%**."*

Chart: Tamil Nadu district map

What it shows (party mode): District polygons colored by **2026 winning party** (aggregated from AC winners in that district).

What it shows (flip mode): Gradient = share of ACs in the district where normalized winner changed 2021→2026.

Script:

"The map is a geographic complement to the mosaic. In party mode, you see which district-level color dominates in 2026. In flip mode, darker districts had more seat changes. Pan and zoom for detail—constituency-level truth is still the 234-tile mosaic on Overview."

Hackathon link: Q1.

Chart: Seats by macro-region (stacked bars)

What it shows: Seat count by **normalized party** within each of six editorial macro-regions (Chennai Metro, North, Central, Kongu, Delta, South). Respects **year toggle**.

Script (2021):

*"With 2021 selected, Chennai Metro is heavily DMK—**29 seats** in this regional split."*

Script (2026):

"With 2026 selected, the stacks show how TVK, DMK, and AIADMK divide each macro-region's seats—useful for a regional segment in the TV show."

Hackathon link: Q1, Q3.

Chart: Flip rate by macro-region

What it shows: Percent of ACs in each macro-region where normalized winner changed (2021 vs 2026). **Not year-toggled**—always the comparison.

Script:

*"This bar chart ranks regions by churn. **Chennai Metro** is at the top—nearly **94%** of constituencies flipped. **Delta** is about **46%**—still high, but lower than the state average of **70%**."*

Hackathon link: Q1, Q2.

Page 4 — Margins (/margins)

Page purpose: How tight races were—**Q6 fragmentation**.

Key insights panel**Script:**

*"Margins tell us competition got closer on average: **103 constituencies** had a 2026 winning margin under **five percentage points**, and only **14 winners** cleared **50%** of valid votes."*

Chart: 2021 vs 2026 margin (scatter)

What it shows: One dot per AC. Typically X = 2021 margin %, Y = 2026 margin % (valid-vote denominator). Points below the diagonal = tighter race in 2026.

Script:

*"Each dot is one constituency. Many dots sit lower on the 2026 axis than on 2021—consistent with the statewide average margin dropping from **11.8%** to **7.7%**. Scroll and zoom to inspect outliers. Hover for AC name."*

Hackathon link: Q6.

Chart: Winner vote share buckets (2026)

What it shows: How many ACs fall into winner share bands: **under 35%**, **35–50%**, **over 50%**.

Bucket	Constituencies
<35%	61
35–50%	159
50%+	14

Script:

*"Most 2026 winners fell between **35 and 50%** of valid votes. **61** won with under **35%**—a plurality. Only **14** crossed **50%**—few landslide-style majorities in this file."*

Hackathon link: Q6.

Table: Closest races

What it shows: Constituencies with the **smallest 2026 winning margin** (winner vs runner-up, % of valid votes).

Script:

"The table lists the knife-edge seats—useful for a short 'closest contests' segment without naming campaigns or causes."

Hackathon link: Q6.

Page 5 — Reserved (/reserved)

Page purpose: Seat outcomes by **GEN / SC / ST** reservation category—descriptive counts only (Q4).

Key insights panel

Script:

“Tamil Nadu has **188 general**, **44 SC**, and **2 ST** constituencies in the master file. Flip rates are similar across GEN and SC—about **70%**—with **31 of 44 SC** seats changing normalized winner.”

Chart: Flip rate by category

What it shows: % of ACs in each reservation type where normalized winner changed 2021→2026.

Category	Flips	Total	Flip %
GEN	131	188	69.7%
SC	31	44	70.5%
ST	1	2	50.0%

Script:

“This chart compares flip **rates** by reservation type—not voter behaviour by community. **SC-reserved** seats flipped at roughly the same rate as general seats in this dataset.”

Hackathon link: Q4.

Table: 2026 seats · party × reserved

What it shows: Exact 2026 seat counts per party in GEN, SC, and ST columns.

Script:

“The matrix is the 2026 seat arithmetic broken down by reservation category—handy for a single slide on how seats distributed without making claims about any group’s voting intent.”

Hackathon link: Q4.

Page 6 — Depth (/depth)

Page purpose: Turnout (2021 only in starter data), ballot size, NOTA—Q5 partial.

Key insights panel

Script:

*"Depth is where we're explicit about data limits: **2026 turnout is blank** in the starter CSV, so turnout charts use **2021 constituency-level turnout**. There is **no booth-level or age data** in the files provided."*

Chart: Average turnout by macro-region (2021)

What it shows: Mean reported **2021 turnout %** per macro-region.

Script:

*"Statewide average turnout in the 2021 file is about **73.4%**. This chart compares macro-regions—useful context if the show opens with 2021 baseline, with a clear label that 2026 turnout isn't in this dataset yet."*

Hackathon link: Q5 (partial).

Chart: Turnout vs 2026 margin (scatter)

What it shows: X = 2021 turnout; Y = 2026 margin. Descriptive correlation only—**do not claim causation**.

Script:

"Each dot pairs 2021 turnout with how tight the 2026 race was. We describe the pattern—we do not say high or low turnout 'caused' a margin."

Chart: Candidates on the 2026 ballot

What it shows: Distribution of **how many candidates** appeared per AC in 2026 (avg **17.2**, max **79**).

Script:

*"2026 ballots were crowded—averaging about **seventeen** candidates per constituency in this file. That's ballot breadth, not booth-level fragmentation."*

Chart: Highest NOTA share (2026)

What it shows: Top constituencies by NOTA as % of valid votes (typically under ~1% in this file).

Script:

*"NOTA stayed a small share statewide; this chart highlights constituencies where it was relatively highest—still under about **one percent** at the top of the list in our processed file."*

Page 6b — Deep insights (/deep)

Page purpose: Structural electoral-science measures applied to the same ECI data. Each index is a standard, published academic metric — no causal claim.

Key insights panel**Script:**

*"Beyond seats and margins, three textbook indices describe how the system itself changed. The Effective Number of Parties — Laakso-Taagepera — rose from **3.74 to 4.29**. Pedersen volatility — net party vote-share churn — is **36**, which the literature flags as very high. Gallagher disproportionality fell from **15.2 to 8.7** — seats track votes more closely than in 2021. These are descriptive measures, not opinions."*

Chart: ENP by region (grouped bars)

What it shows: Effective number of parties per macro-region, 2021 vs 2026.

Script:

*"Every region is more fragmented in 2026 except Chennai Metro, which stayed near 3.3 — same number of effective parties, but Pedersen volatility tells us the parties themselves changed. South Tamil Nadu is most fragmented at **4.6**."*

Chart: Pedersen volatility by region (horizontal bars)

What it shows: Half the sum of absolute vote-share changes per region.

Script:

*"Pedersen is highest in Chennai Metro at **49** — the largest party-share churn between elections in the file. Central and Kongu sit in the mid-thirties. Twenty-plus is considered very high in academic literature."*

Chart: Swing heatmap (party × region)

What it shows: Vote-share change in percentage points, party rows × region columns. Diverging palette: green = gain, red = loss.

Script:

"The heatmap traces party gains and losses by region. We describe the colors — green means a party's vote share went up in that region, red means down. We do not say where one party's votes 'went' — votes don't literally transfer between parties."

Chart: Anti-incumbency by region

What it shows: Share of seats where the 2021 winning party did **not** win in 2026.

Script:

*"Anti-incumbency is a descriptive label for seat-level non-retention. Statewide it's **69.7%**. Chennai Metro is highest at **94%**, Delta is lowest at **46%**. The chart counts seat outcomes; it does not infer voter motive."*

Chart: Representation gap (vote share vs seat share, 2026)

What it shows: Grouped horizontal bars per party — vote share % vs seat share %.

Script:

*"First-past-the-post amplifies the leading party. TVK won **35%** of the vote and **46%** of the seats — an **eleven-point** amplification. Smaller parties like BJP got under-represented relative to votes. Gallagher LSq summarises this gap into one number, and it's lower than in 2021."*

Chart: Race competitiveness (buckets)

What it shows: Combined top-2 vote share bucket: multi-cornered (<70%), three-way (70–85%), two-party (>85%).

Script:

*"**132 of 234** races in 2026 were multi-cornered — top two candidates combined for less than 70% of the valid votes. Only **four** were two-party contests. That structural shift sits behind the margin collapse on the Margins page."*

Table: District churn

What it shows: Districts sorted by share of ACs that flipped, with a "full sweep" badge for districts where every AC changed normalised winner.

Script:

"The table lists every district by churn. Full-sweep districts are flagged — useful for a regional B-roll segment without naming campaigns."

Hackathon link: Goes beyond the six required questions — structural electoral-science context.

Page 7 — Explorer (</explorer>)

Page purpose: Analyst view—filter ACs and see **filtered** tallies, margin scatter, and table (not statewide totals).

Filter bar

What it does: Region, reserved category, 2026 winner party, flip-only, max margin slider.

Script:

*"Explorer is for follow-up questions—for example, 'show only flipped seats in Kongu with margin under five percent.' The KPI strip and seat tally **recalculate for the filter**, unlike Overview which is always statewide unless you use region chips on the mosaic."*

Mini KPIs: Matching ACs / Flips in set / Avg margin 2026

Script:

"These three numbers summarize whatever filter you applied—great for rehearsing a regional deep-dive segment."

Chart: Filtered seat tally

What it shows: 2026 seat counts **within the current filter**.

Script:

"After filtering, the bar chart answers 'who won among these constituencies only?'"

Chart: Margin shift (filtered scatter)

What it shows: Same 2021 vs 2026 margin view, subset of ACs. **Click a dot** → detail card.

Script:

"Click any point to pull up that constituency's 2021 and 2026 winners and margins for on-air fact-checking."

Table: Constituency list

What it shows: AC number, name, 2021 winner, 2026 winner, 2026 margin. Row click → detail.

Script:

"The table is the searchable backbone—use it if the producer asks for a specific seat during recording."

Page 8 — Sources ([/methods](#))

Page purpose: Sources, metric definitions, limitations, reuse notes—for closing the video.

Script:

"All metrics come from ECI starter CSVs joined on assembly constituency number. Winners use valid votes excluding NOTA. Margins are winner minus runner-up, divided by valid votes. Flips mean normalized 2021 winner differs from normalized 2026 winner. Party labels are normalized via our mapping file—raw ECI strings would change counts slightly. We do not use exit polls, news, or social media."

Research threads — one-line bridges (internal; do not read Q labels on air)

Thread	One sentence for video
Geographic patterns	Flips clustered highest in Chennai Metro (93.8%) , lowest in Delta (45.5%) ; regional stacks show how seats split in 2021 vs 2026.
Seat flips	163 ACs (69.7%) changed winner; largest flow DMK 2021 → TVK 2026 (65 seats) .
Vote share / TVK	TVK: 108 seats, ~35% statewide valid vote share in 2026; DMK had ~38% vote share and 133 seats in 2021.
Reserved seats	188 GEN / 44 SC / 2 ST ; SC flip rate 70.5% vs GEN 69.7% —similar churn, different seat totals.
Turnout	2021 avg ~73.4% in CSV; 2026 turnout not provided in starter files.
Margins	Avg margin 11.8% → 7.7% ; 61 winners under 35% share; 103 seats under 5 pt margin.
Electoral arithmetic	ENP 3.74 → 4.29 ; Pedersen 36 ; Gallagher LSq 15.2 → 8.7 ; 132 of 234 multi-cornered races.

Editorial recommendation (closing — on camera or deck)

Template (customize order/timing):

"For a sixty-minute neutral special, I'd recommend: (1) statewide arithmetic—108 TVK seats, 163 flips, margin drop [Statewide, 10 min]; (2) regional map and flip bars for Chennai vs Delta [Geography, 15 min]; (3) churn and close races—margins under five points and the seat-flow Sankey [Margins + Seat flows, 10 min]; (4) sources and caveats—no 2026 turnout in file, normalization rules [Sources, 5 min]. Use the constituency explorer if producers need a specific seat during the show. Keep vote share and TVK in segment one so seats and votes are not conflated."

Limitations (always say once)

*"This analysis is constituency-level ECI data only. **2026 turnout is missing** in the provided CSV. There are **no polling-booth or voter-age tables**. Party names are **normalized** for comparison. Maps approximate district boundaries. Nothing here predicts future elections or explains voter motivation."*

Quick reference — headline numbers

Metric	Value
Total ACs	234
TVK seats 2026	108
DMK seats 2021	133
Flips (normalized)	163 (69.7%)
Avg margin 2021 / 2026	11.8% / 7.7%
Winners <35% vote share 2026	61
Winners >50% vote share 2026	14
Margins <5 pp 2026	103
TVK vote share 2026	~35.1%
DMK vote share 2021	~38.0%
Largest Sankey flow	DMK → TVK, 65 seats
DMK retention	40 / 133 (30.1%)
Effective Number of Parties	3.74 → 4.29 (Laakso-Taagepera)
Pedersen volatility	36.4 (literature: 20+ = high)
Gallagher LSq	15.2 → 8.7 (lower = more proportional)
Multi-cornered races (top-2 < 70%)	132 of 234

Related docs

- [docs/METRIC_DEFINITIONS.md](#) — formulas
- [docs/DASHBOARD_QA.md](#) — what to click when testing
- [HACKATHON_EXECUTION_PLAN.md](#) — deliverables and rubric
- Dashboard live insights: </api/insights?year=2026>

Regenerate numbers after rebuilding processed data.

Visual & Reference Master

Visual Design & Reference Master Guide

Current submission stack (May 2026): React (`web/`) + FastAPI (`api/`) + ECharts. Run `uvicorn api.main:app --port 8000` (after `cd web && npm run build`) or `npm run dev` on `:5173`. Sections that mention **Streamlit** or `dashboard/app.py` are **design references only** — not the shipped RPC dashboard.

Deep insights page now live (`/deep`): Effective Number of Parties (Laakso–Taagepera), Pedersen volatility, Gallagher LSq disproportionality, vote-share swing heatmap, anti-incumbency, representation gap, race competitiveness, district churn. All descriptive, ECI-only. Definitions in `docs/METRIC_DEFINITIONS.md`.

Purpose: Single reference for building a **world-class, neutral, creative** Tamil Nadu 2026 election story — not limited to Codebasics rules. Use this alongside `HACKATHON_EXECUTION_PLAN.md` for submission logistics.

How to use: Pick **one headline story** → choose **3 chart types** from Section 8 → apply **accessibility + ethics** (Sections 9–10) → implement with **tech stack** (Section 11) → validate against **data sources** (Section 12).

1. Reference ecosystem (who to learn from)

```

flowchart TB
  subgraph challenge [Your deliverables]
    V[Video pitch]
    D[Stakeholder deck]
    G[GitHub notebook]
    DB[Interactive dashboard]
  end

  subgraph global [Global newsroom patterns]
    BBC[BBC India 2024 maps]
    NPR[NPR tile maps + live results]
    FT[FiveThirtyEight / 538 flows]
    AP[AP VoteCast Sankey]
    Vox[Vox cartogram critique]
  end

  subgraph india [India public examples]
    Elexa[News18 Elexa]
    Quint[Quint Data Studio]
    Hindu[The Hindu infographics]
    IIT[IIT Bombay Election Explorer]
    DM[Datameet maps + data]
  end

  subgraph creative [Creative / scrolllytelling]
    Pudding[The Pudding visual essays]
    IIB[Information is Beautiful Awards]
  end

  subgraph standards [Standards and ethics]
    RSS[RSS DataVis Guide]
    EU[EU Data Viz Accessibility]
    APeth[AP factual reporting]
    RPC[Codebasics RPC neutrality]
  end

  global --> DB
  india --> DB
  creative --> V
  creative --> D
  standards --> challenge
  RPC --> challenge

```

2. Codebasics RPC (mandatory constraints)

Item	Requirement
Deadline	28 May 2026, 11:59 PM IST
Scoring	~70% storytelling (video + deck), ~30% data hygiene
Data	ECI public + starter CSVs only
Tone	No causal claims, predictions, party praise/criticism
Visuals	No leader photos, party symbols
Deliverables	Video 5–7 min → Deck 8–10 slides → GitHub → Dashboard optional
Research	Pick 3 of 6 questions; weave one story

Full detail: [HACKATHON_EXECUTION_PLAN.md](#), [How_to_Submit.docx](#), [metadata.txt](#).

3. Global newsroom dashboards — patterns to adopt

3.1 BBC News — India 2024 election results

- **URL:** [BBC India 2024 results \(interactive\)](#)
- **Patterns:** Constituency-level drill-down; **year-over-year compare** (2024 vs 2019); battleground highlights; search; map scaled for **readability over geographic truth**
- **Steal for TN:** Compare **2026 vs 2021** at AC level; default to **regional summary** then drill to constituency

3.2 NPR — Election results & tile maps

- **Live results:** [NPR 2024 election apps](#)
- **Tile maps:** [NPR hex/tile map methodology](#)
- **Patterns:** Don't over-color incomplete counts; **uniform tiles** so rural ACs aren't visually erased; sidebar filters
- **Steal for TN:** **234-tile mosaic** grouped by 6 editorial regions (not geographic choropleth if no clean shapefile)

3.3 Associated Press — AP VoteCast

- **URL:** [AP VoteCast 2024 \(IIB showcase\)](#)
- **Patterns:** **Sankey** for flows between groups; scan path for casual users; deep tables for power users
- **Steal for TN:** **2021 winner party → 2026 winner party** Sankey (seat flows, not voter demographics)

3.4 FiveThirtyEight / 538

- **Forecast UX:** [538 2020 forecast \(IIB\)](#)
- **Demographics:** [Swing-O-Matic style tools](#)
- **Patterns:** Uncertainty as first-class citizen; plain-language “robotext”; accessibility audits; **avoid** implying prediction for RPC
- **Steal for TN:** **Margin distributions** and “how fragmented is the mandate” — descriptive only, no forecast

3.5 Vox & National Geographic — map literacy

- **Vox:** [Why election maps mislead](#)
- **NatGeo:** [Cartogram alternatives](#)
- **Lesson:** Choropleths overweight land area; cartograms cost geography literacy; **tile grids** are the compromise
- **Steal for TN:** If you use a map, add **bar chart of same data** side-by-side (dual encoding)

3.6 ICA research — map type affects accuracy

- **Source:** Copernicus ICA work on election map types (choropleth vs cartogram vs block maps)
 - **Lesson:** Saturated reds/blues increase perceived partisanship; **desaturate**; accent only key views
 - **Steal for TN:** Use **muted Okabe–Ito** palette; never saffron/green dyads
-

4. India — public dashboards & data hubs

Source	URL	What to study
ECI results portal	https://results.eci.gov.in/	Official 2026 TN live results; turnout if missing in CSV
ECI statistical reports	https://www.eci.gov.in/statistical-reports	PDF validation, historical context
TN CEO	https://elections.tn.gov.in/	State Form-20, master lists
data.gov.in (ECI)	https://data.gov.in	Downloadable CSV/Excel
News18 Elexa	https://www.news18.com/elections/analytics-center/	Filters, grid/list/map, search
Quint Data Studio	https://www.thequint.com/quintlab/	Live seat maps, constituency explorer
The Hindu infographics	e.g. Bihar 2025 results pages	Clean small-multiples, filterable charts
IIT Bombay Election Explorer	https://info-design-lab.github.io/General-Election-Explorer/	Cartogram + map; ECI + Datameet
Parliament 2009–2024 explorer	https://data-analytics.github.io/Election_Data_2024/	Multi-cycle compare UX
IndiaVotes	https://indiavotes.com	Cross-check constituency numbers
Datameet election data	https://github.com/datameet/india-election-data	Open historical datasets
Datameet AC shapefiles	https://github.com/datameet/maps/tree/master/assembly-constituencies	Choropleth (verify delimitation caveats)
Trivedi Centre	Ashoka / TCPD (2021 TN source in metadata)	Academic cleaning standards

TN-specific note: Constituency boundaries on Datameet may be **pre-delimitation** for some states — document in limitations if used.

5. Creative & award-winning visual journalism

5.1 The Pudding — visual essays

- **Method:** [How Pudding structures visual essays](#)
- **Scrollytelling:** [Scrollytelling examples](#)
- **Patterns:** Pinned chart + scrolling narrative; sparse words; data carries conclusion
- **Steal for RPC: Video** = spoken scrollytelling; deck slide 1 = “pinned” headline chart

5.2 Information is Beautiful Awards

- Browse: <https://www.informationisbeautifulawards.com/> (Current Affairs & Politics)
- **Steal:** One **hero visual** (Sankey or mosaic) that works without reading body text

5.3 Other high-impact formats (industry)

Format	Best for	Example use in TN RPC
Sankey / alluvial	Party seat flows 2021→2026	Q2 Flips
Tile / waffle grid	234 ACs equal visual weight	Q1 Geographic + explorer
Sunburst / treemap	Region → party hierarchy	Regional seat composition
Slope / bump chart	Rank or metric change by entity	Margin 2021 vs 2026 by AC sample
Beeswarm / strip	Distribution of margins	Q6 fragmentation
Marimekko	Vote share composition by region	Q3 TVK footprint (descriptive)
Chord diagram	Flow matrix (use sparingly — hard to read)	Alternative to Sankey
Parallel coordinates	Multi-metric AC profiles	Power-user dashboard tab
Small multiples	6 regions same chart type	Q1
Dumbbell chart	Two-year comparison	Turnout delta (if scraped)

6. Accessibility & inclusive design (required for “best” work)

Standard	URL	Apply
EU Data Visualization Guide	https://data.europa.eu/apps/data-visualisation-guide/	Contrast $\geq 4.5:1$ text; don't use color alone
RSS Best Practices	https://royal-statistical-society.github.io/datavisguide/	Direct labels, alt text, chart choice
Urban Institute — Do No Harm	https://www.urban.org/research/publication/do-no-harm-guide	Center accessibility from day one
CT Open Data guidelines	https://ctopendata.github.io/data-visualization-guidelines/	Audience, goals, tables alongside charts
Okabe-Ito palette	Standard colorblind-safe set	Default party-agnostic colors
WCAG 2.x	W3C	Keyboard nav for web dashboard; semantic HTML

Checklist before ship:

- ☐ No red–green only encoding
- ☐ Labels on lines/bars (not legend-only)
- ☐ Tooltips + **data table** export for each main chart
- ☐ Alt text / short description for deck stills
- ☐ Font size ≥ 12 pt equivalent on TV-safe deck charts

7. Neutral reporting & ethics (aligns with RPC disqualification rules)

Source	Principle
AP Elections	Report facts; explain methodology; no “likely winner” without math path
ACE Project (media ethics)	Separate fact from opinion; balance; label commentary
RPC brief	No causal “why party won”; no predictions; ECI-only

Safe chart title templates:

- ☒ “108 constituencies had TVK as leading candidate in published results”
- ☒ “TVK destroyed DMK in the south”

-  "Average winning margin was 7.7% in 2026 vs 11.8% in 2021 (valid votes, excl. NOTA)"
-  "Voters rejected the ruling party"

8. Chart playbook → TN 2026 research questions

RPC question	Primary chart	Secondary	Inspiration
Q1 Geographic	Regional stacked bars OR tile mosaic by region	Small multiples map	BBC, NPR tiles
Q2 Flips	Sankey (2021 party → 2026 party)	Highlight table of flipped ACs	AP VoteCast
Q3 TVK vote share	Marimekko or grouped bars by region	Dot plot statewide shares	538 composition charts
Q4 Reserved seats	Faceted bars GEN vs SC vs ST	Same metrics, side-by-side years	Hindu small-multiples
Q5 Turnout	Dumbbell top-20 ACs	Choropleth if turnout scraped	ECI portal + Quint live maps
Q6 Margins	Beeswarm/box/violin	KPI: % seats winner <35% share	FiveThirtyEight uncertainty framing

Recommended "creative hero" trio (data-backed):

1. **234-cell tile mosaic** (2026 winner, filter by region/flip)
2. **Sankey** seat flows 2021→2026
3. **Margin beeswarm** 2021 vs 2026 overlay

9. Tech stack decision matrix

Tool	Strengths	Weaknesses	Best for
Streamlit	Fast Python, filters, Plotly embed	Custom design limits	Hackathon dashboard + demo
Plotly / Dash	Sankey, interactivity, export	Dash learning curve	Sankey + drill-down
Tableau Public	Polish, sharing gallery	Repro on GitHub harder	Impressive static/interactive publish
Power BI	Corporate deck alignment	Less "creative web" feel	If reviewer expects Microsoft stack
D3.js / Observable	Maximum creativity	Time-intensive	Portfolio-grade custom viz
Flourish	Animated bars, Sankey templates	Hosted dependency	Quick hero GIF for LinkedIn
Jupyter + Matplotlib	Reproducibility	Less interactive	Notebook grading
GeoPandas + Folium/Plotly	Real maps	Boundary QA burden	Only with Datameet + caveats

Recommended stack for "creative + submittable":

```

pandas → processed CSVs
plotly → Sankey, mosaic, beeswarm
streamlit → multipage app (st.navigation)
kaleido → PNG exports for deck
jupyter → reproducibility

```

10. Dashboard information architecture (BBC + NPR + News18 hybrid)

10.1 Page structure (Streamlit multipage)

Page	User question	Key widgets
Overview	"What happened statewide?"	KPI row, headline, editorial rec card
Flow	"How did seats move?"	Sankey, flip count
Regions	"Where?"	Tile mosaic + regional bars
Fragmentation	"How close were races?"	Margin beeswarm, <35% table
Explorer	"Tell me about one AC"	Search <code>ac_number</code> , candidate table
Methods	"Can I trust this?"	Definitions, sources, limitations

10.2 UX best practices

- **Streamlit:** `st.set_page_config(layout="wide")`, `@st.cache_data`, filters in sidebar ([Streamlit multipage docs](#))
- **NPR:** Equal visual weight per constituency in grids
- **BBC:** Year toggle 2021 | 2026 everywhere
- **News18:** Search + sort on constituency table
- **Do not:** Auto-play sound; flashing colors; 3D pie charts

10.3 Visual design system (neutral)

```

Background:  #0f1419 (dark) or #f8fafc (light) – pick one theme, stay consistent
Text:        #e2e8f0 / #1e293b
Accent:      #3b82f6 (navigation only – not party-coded)
Party series: Okabe-Ito – assign arbitrarily by alphabet, document in README
Font:        Inter, Source Sans 3, or system-ui
Chart height: 480–600px for hero; 320px for small multiples

```

11. Data pipeline (all sources)

```

Starter pack (ac_number join)
  tn_2021_results.csv
  tn_2026_results.csv
  constituency_master.csv

Optional enrichment
  ECI results.eci.gov.in → 2026 turnout (blank in starter CSV)
  eci.gov.in/statistical-reports → audit PDFs
  datameet/maps → AC boundaries (with limitation footnote)
  indiavotes.com → spot checks

Derived tables (you create)
  ac_summary_2021.csv
  ac_summary_2026.csv
  ac_comparison.csv
  party_normalize.csv
  sankey_edges.csv

```

Metrics: See `docs/METRIC_DEFINITIONS.md` (create when building) — valid votes excl. NOTA, margin%, flip flag.

12. Creative build blueprint (end-to-end)

Phase A — Data (Day 1–2)

1. Copy raw → `data/raw/`
2. Build `party_normalize.csv` (top 30 raw labels)
3. Export `ac_comparison.csv` (234 rows)
4. Validate 2 ACs manually against ECI

Phase B — Hero visuals (Day 3–5)

1. **C1** Tile mosaic — Plotly scatter grid or custom CSS grid in Streamlit
2. **C2** Sankey — `go.Sankey` (Plotly docs)
3. **C3** Margin beeswarm — `px.strip` or violin by year
4. Export 1920×1080 PNG → `outputs/charts/`

Phase C — Dashboard (Day 5–7)

1. `streamlit run dashboard/app.py`
2. Multipage navigation + sidebar filters (region, reserved, flip)
3. `st.query_params` for shareable AC links (optional)

Phase D — Story package (Day 7–10)

1. Deck: hero = C1, slides 2–4 = C2–C3 + one regional bar
2. Video: screen-record dashboard + face cam
3. README: reproduction + neutrality disclaimer

Phase E — Submit (Day 10–12)

1. GitHub public
2. YouTube unlisted
3. LinkedIn (tags per `How_to_Submit.docx`)
4. RPC portal URL

13. Master submission checklist (everything)

Content quality

- ☐ One sentence headline works standalone
- ☐ 3 research questions answered with depth
- ☐ Editorial recommendation for 60-min show
- ☐ Limitations slide (Form-20, normalization, turnout)

Visual quality

- ☐ Hero visual understandable in 5 seconds
- ☐ 3 distinct chart types (not 3 bar charts)
- ☐ Accessibility checklist (Section 6)
- ☐ Neutrality tone audit (Section 7)

Technical

- ☐ Notebook runs clean env
- ☐ Dashboard runs via documented command
- ☐ Numbers match between deck, video, dashboard

Distribution

- ☐ Public GitHub
- ☐ Video link
- ☐ LinkedIn: `@codebasics`, Hemanand Vadivel, Dhaval Patel, Govt of Tamil Nadu
- ☐ Hashtags: `#ResumeProjectChallenge #Codebasics #DataAnalytics #TamilNaduElection2026`
- ☐ RPC page submission

14. Curated bibliography (bookmark list)

Maps & geography

- NPR tile maps: <https://blog.apps.npr.org/2015/05/11/hex-tile-maps>
- Vox on misleading maps: <https://www.vox.com/2016/6/2/11828628/election-maps-hard>
- Datameet AC maps: <https://github.com/datameet/maps/tree/master/assembly-constituencies>

Interactivity & charts

- Plotly Sankey: <https://plotly.com/python/sankey-diagram/>
- Plotly Sankey deep dive: <https://plotly.com/blog/sankey-diagrams>
- Python graph gallery Sankey: <https://python-graph-gallery.com/sankey-diagram-with-python-and-plotly/>

Journalism & storytelling

- Pudding structure: <https://www.storybench.org/pudding-structures-stories-visual-essays/>
- AP election role: <https://www.ap.org/elections/our-role/how-we-call-races>

Accessibility

- RSS styling guide: <https://royal-statistical-society.github.io/datavisguide/docs/styling.html>
- EU accessible palettes: <https://data.europa.eu/apps/data-visualisation-guide/accessible-colour-palettes>

India data

- ECI results: <https://results.eci.gov.in/>
- IIT Election Explorer: <https://info-design-lab.github.io/General-Election-Explorer/>
- Datameet india-election-data: <https://github.com/datameet/india-election-data>

Tools

- Streamlit multipage: <https://docs.streamlit.io/develop/concepts/multipage-apps/page-and-navigation>
- Information is Beautiful Awards: <https://www.informationisbeautifulawards.com/>

This project

- `HACKATHON_EXECUTION_PLAN.md`
- `input_files_for_participants_rpc/`

15. What “most creative” means for this RPC

Creativity $\neq$ more chart types. It means:

1. **One unforgettable hero** (tile mosaic or Sankey) that reviewers remember after the video

2. **Clear narrative arc** in the dashboard (Overview → Why it matters → Proof → Explorer)
3. **Dual encoding** (mosaic + table, map + bar) so accessibility and trust increase
4. **Honest limitations** surfaced in UI, not hidden in README footnotes
5. **Production polish** — consistent theme, no clutter, fast filters

Avoid: 3D pies, party-flag colors, leader images, causal headlines, prediction language.

Document version: 1.0 — consolidates global, India, creative, ethical, and RPC-specific references.

Agent Instructions

Agent instructions — TN Election RPC

For any work on this hackathon, use:

1. `HACKATHON_EXECUTION_PLAN.md` — Internal submission checklist. Dashboard opens on **Editorial brief** (`/`).
2. `docs/VIDEO_NARRATION_SCRIPT.md` — per-page and per-chart narration for video recording. 2b. `docs/CHART_LAYOUT_AUDIT.md` — per-chart clipping / margin checklist.
3. `docs/DASHBOARD_QA.md` — routes, E2E, manual QA.
4. `docs/DASHBOARD_REUSE.md` — reuse pipeline for another election.
5. `docs/VISUAL_AND_REFERENCE_MASTER.md` — creative reference (Streamlit sections are bibliography only).
6. `docs/METRIC_DEFINITIONS.md` — winner, margin, flip, valid votes, plus advanced indices (ENP, Pedersen, Gallagher LSq, representation gap) and constitutional context.
7. `deck/TN_2026_AtliQ_Briefing.pptx` — 18 slides; rebuild via `python scripts/build_presentation.py`.
8. `src/advanced_metrics.py` — electoral-science indices powering the `/deep` page and `/api/advanced/*` endpoints.
9. `.cursor/rules/tn-election-rpc.mdc` — always-on constraints.

Starter data lives under `input_files_for_participants_rpc/`.